Topological band structure theory

Today’s learning outcomes

- Explain the phenomenology of the quantum Hall state
- Explain why topological states of matter show quantum phenomena resilient to perturbations
- Provide examples of how non-trivial topological invariants give rise to gapless surface excitations

Today’s plan

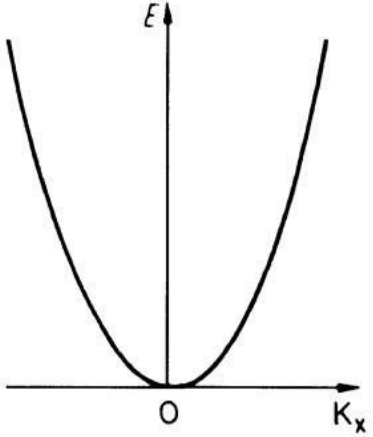
- Linear response and the Hall conductivity
- The quantum Hall effect and its edge states
- The concept of topology in physics
- A minimal model for a topological insulator
- Topology beyond electrons

A reminder from session #3

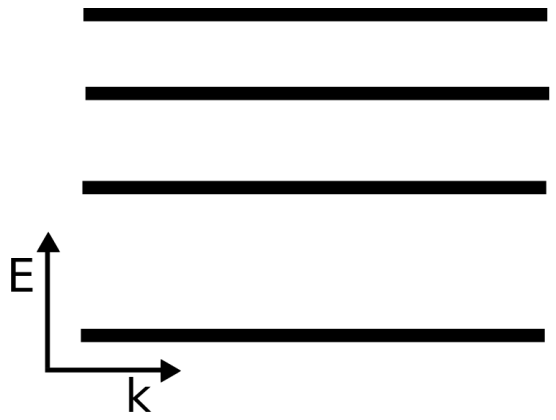
Hamiltonians with translational symmetry can be diagonalized using Bloch's theorem, yielding a band-structure

$$H = \sum_{\alpha,k} \epsilon_{\alpha,k} \Psi^{\dagger}_{\alpha,k} \Psi_{\alpha,k}$$

Parabolic band

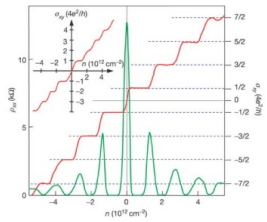


Flat bands



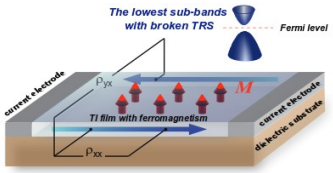
Topological quantum materials

Quantum Hall effect



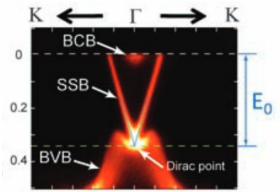
graphene

Quantum Anomalous Hall effect



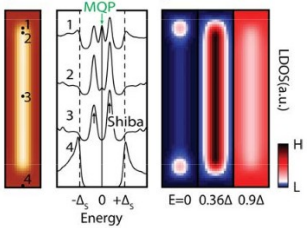
Cr-doped $(Bi,Sb)_2Te_3$

Quantum Spin Hall effect



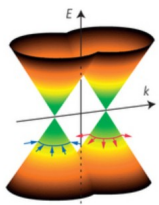
Bi_2Se_3

Topological superconductor



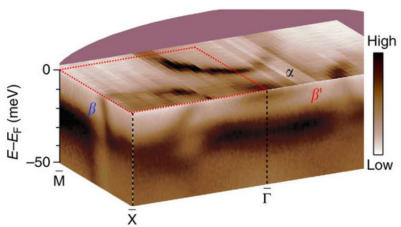
Fe@Pb

Weyl semimetal



TaAs

Topological Kondo insulator



SmB_6

Many different topological states in nature

Linear response theory and the Hall conductivity

Response in materials



In general, for a certain observable *A* we will have

$$A(x, x', t, t')$$

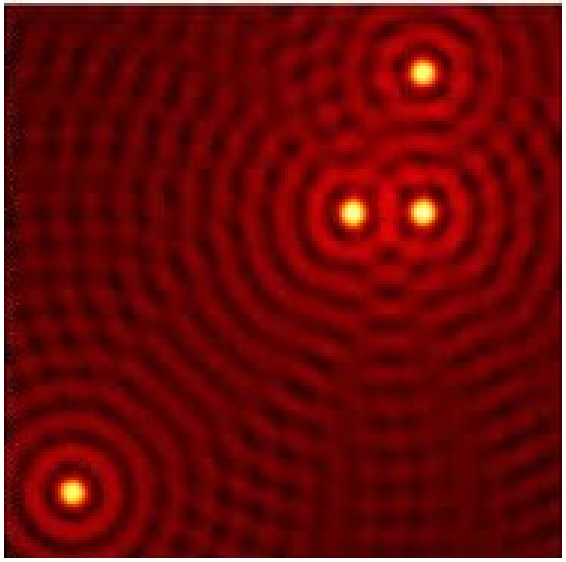
Which kind of perturbation?

→ Charge (dielectric and conductivity)
→ Spin fluctuation (magnetic susceptibility)
→ Heat (thermal conductance)

Types of responses

Impurities in a metal

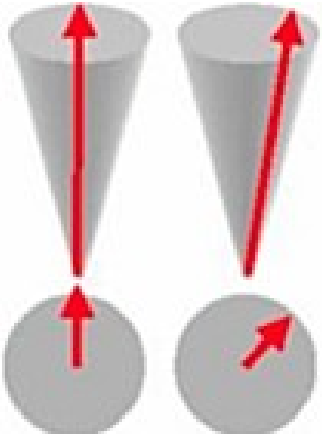
Friedel oscillation

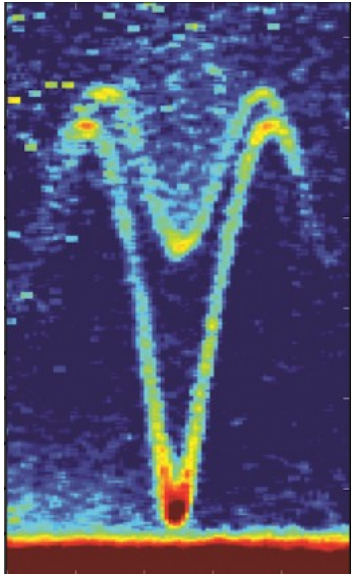


Information on the Fermi surface

Neutron scattering

(Inelastic neutron-matter process)



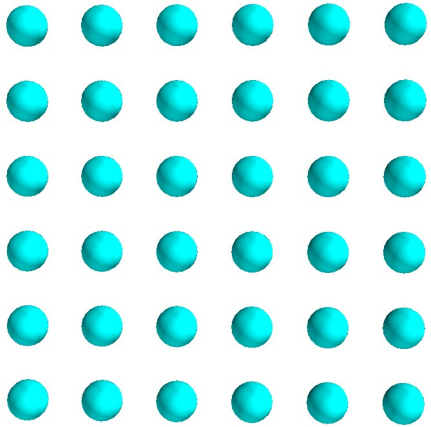


Information on magnetic excitations

Unperturbed and perturbed Hamiltonian

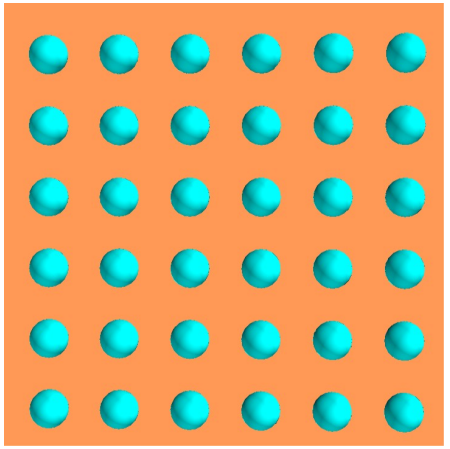
Unperturbed Hamiltonian

$$H_0$$



Perturbed Hamiltonian

$$H = H_0 + \lambda V$$



Can we infer the properties of a perturbed Hamiltonian from the unperturbed one?

Perturbation theory

Take a Hamiltonian that includes a perturbation V

$$H(\lambda) = H_0 + \lambda V$$

Every “well behaved” function is linear for small argument

Energy	***Observables***	***Wavefunctions***
$E_{GS}(\lambda)$	$\langle A \rangle(\lambda)$	$\lvert \Psi_k \rangle(\lambda)$

We will use a Taylor expansion in terms of the parameter λ

Observables in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Full Hamiltonian

Unperturbed Hamiltonian

Perturbation

For any observable

$$A(\lambda) \approx A(\lambda = 0) + \lambda\chi$$

Full value

Unperturbed value

Contribution from the perturbation

Response of the system

$$\frac{\partial A}{\partial \lambda} \equiv \chi$$

Eigenfunctions in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Eigenfunctions after the perturbation

$$|\Psi_i\rangle(\lambda) = |\Psi_i\rangle(\lambda = 0) + \lambda \sum_{j \neq i} \alpha_j |\Psi_j\rangle(\lambda = 0)$$

Eigenfunction after the perturbation

Eigenfunctions before the perturbation

Eigenfunctions in linear response theory

$$H(\lambda) = H_0 + \lambda V$$

Correction to the eigenfunctions to first order

$$|\Psi_i\rangle(\lambda) = |\Psi_i^0\rangle + \lambda \sum_{j \neq i} \alpha_j |\Psi_j^0\rangle$$

Coefficients

$$\alpha_j = \frac{\langle \Psi_i^0 | V | \Psi_j^0 \rangle}{\epsilon_i^0 - \epsilon_j^0}$$

Notation

$$|\Psi_i^0\rangle \equiv |\Psi_i\rangle(\lambda = 0)$$

$$H_0 |\Psi_i^0\rangle = \epsilon_i^0 |\Psi_i^0\rangle$$

The perturbed eigenfunctions can be expressed in terms of the unperturbed ones

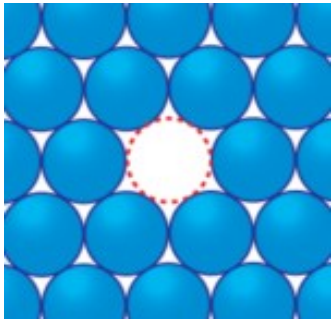
Breakdown of perturbation theory

$$H(\lambda) = H_0 + \lambda V$$

Is any full many-body ground state perturbative in terms of λ ?

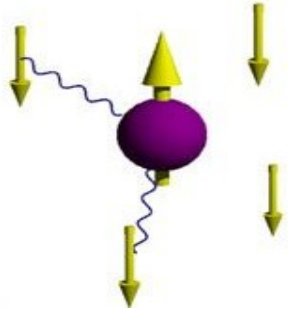
No! Certain quantum states are intrinsically non-perturbative

Impurity in a metal (Orthogonality catastrophe)



The many-body ground state Is orthogonal to the original

Quantum spin in a metal (Kondo effect)



The Kondo temperature is non-perturbative

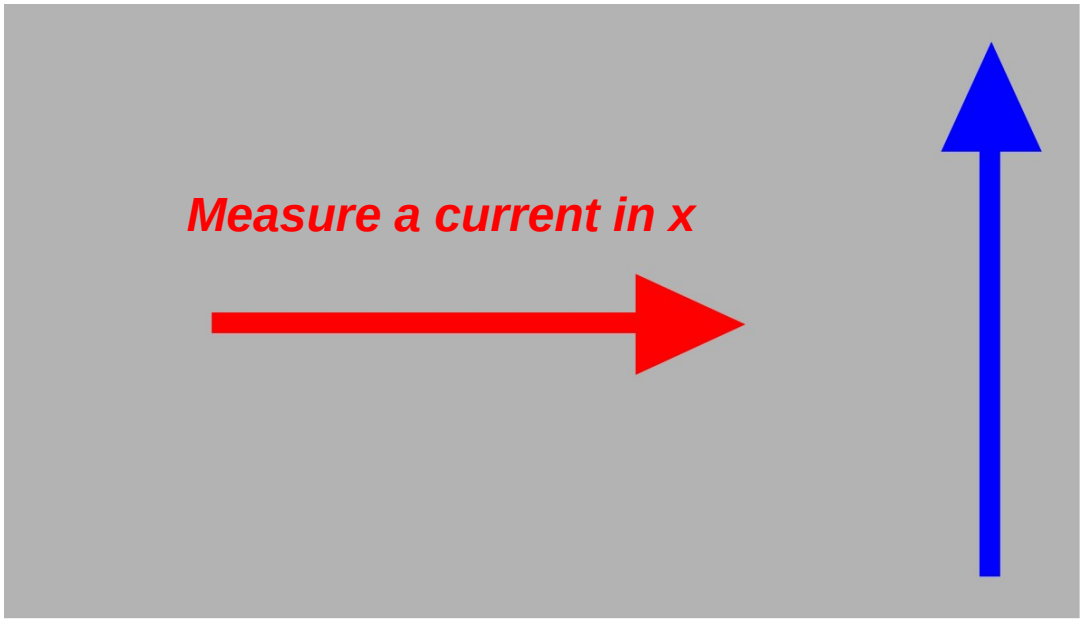
Superconductivity (BCS limit)



The superconducting gap is non-perturbative

The transverse conductivity

Take a two-dimensional material



$$J_x = \sigma_{xy} V_y$$

Full Hamiltonian

$$H = H_0 + \lambda V$$

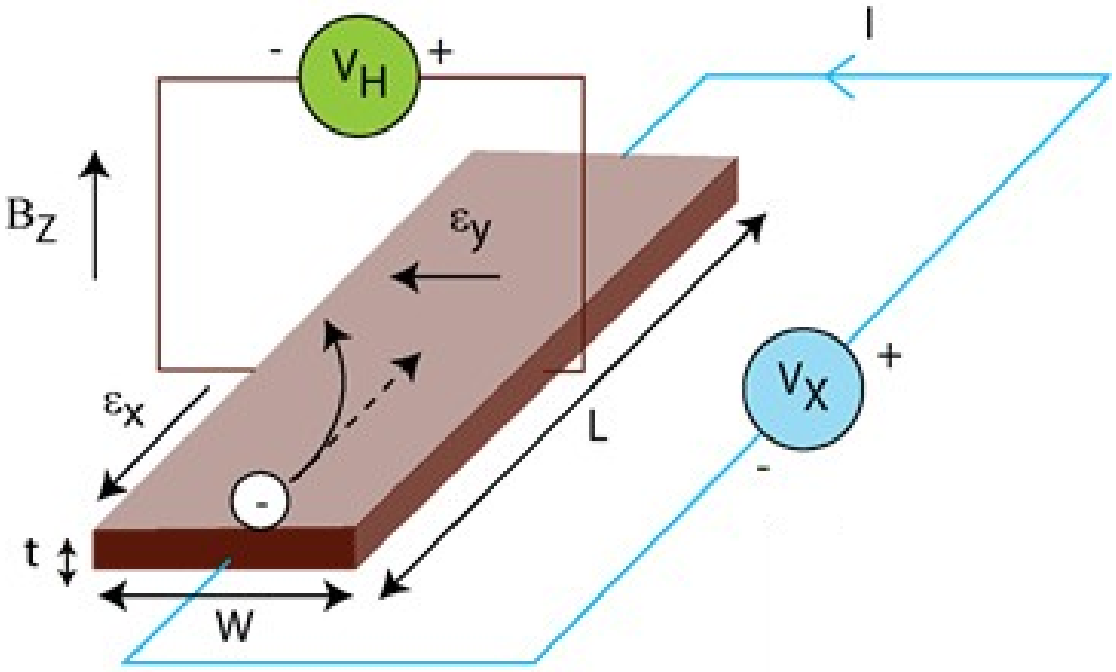
Perturbation

$$V \sim y \sim i\partial_{k_y}$$

Measure

$$J_x \sim \langle \partial H / \partial k_x \rangle$$

The Hall effect



$$J_x = \sigma_{xy} V_y$$

Hall conductivity

Measure the current perpendicular to a voltage

Linear response for transverse current

$$J_x = \sigma_{xy} V_y$$

The Hall conductivity is obtained as $\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k}$

with $\Omega_\alpha(\mathbf{k}) = i \sum_{\beta \neq \alpha} \frac{\langle \Psi_\alpha | \partial H/\partial k_x | \Psi_\beta \rangle \langle \Psi_\beta | \partial H/\partial k_y | \Psi_\alpha \rangle}{(\epsilon_\alpha - \epsilon_\beta)^2} - \alpha \leftrightarrow \beta$

Berry curvature of a band

Expression coming from perturbation theory

The Hall conductivity

The Hall conductivity is obtained as $\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k}$

Using $\langle \Psi_\alpha | \partial H / \partial k_\mu | \Psi_\beta \rangle = \langle \partial_{k_\mu} \Psi_\alpha | \Psi_\beta \rangle (\epsilon_\alpha - \epsilon_\beta)$

the Hall conductivity can be expressed in terms of

Berry curvature

$$\Omega_\alpha = \partial_{k_x} A_y^\alpha - \partial_{k_y} A_x^\alpha$$

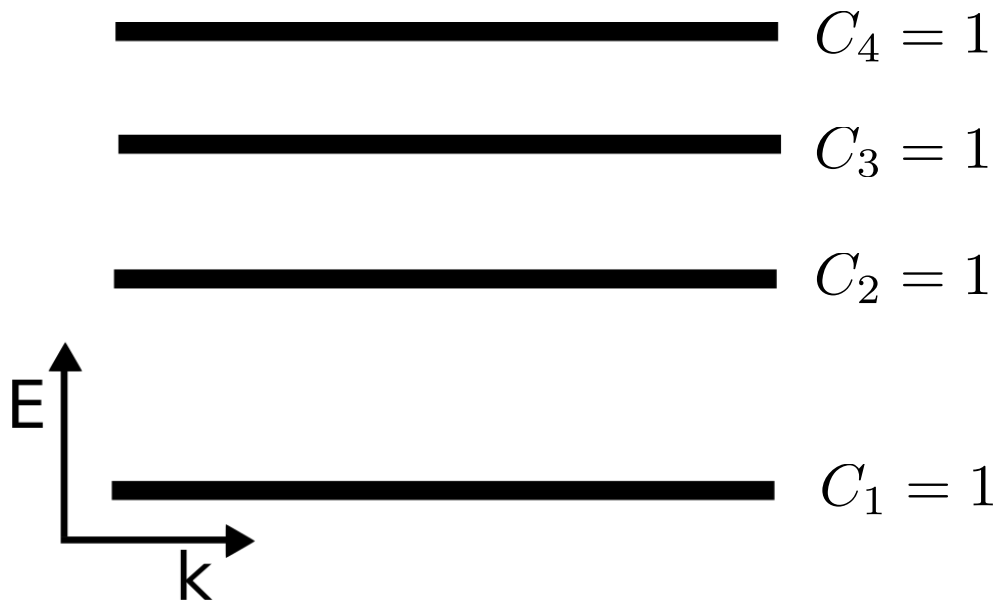
Berry connection

$$A_\mu^\alpha = i \langle \partial_{k_\mu} \Psi_\alpha | \Psi_\alpha \rangle$$

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$ ← Chern number

Chern numbers in the quantum Hall state

Band-structure in the quantum Hall state



Hall conductivity

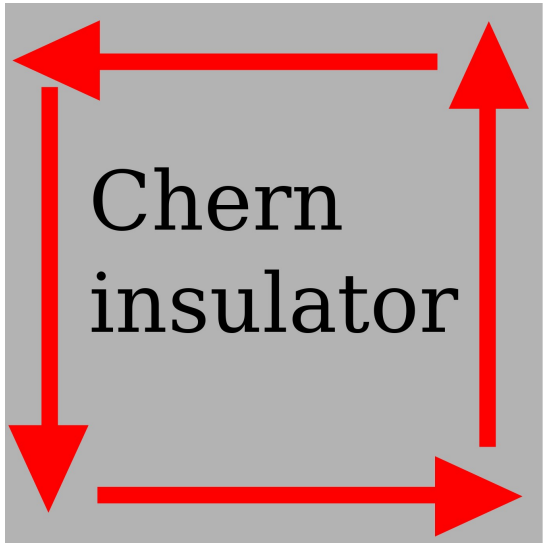
$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$

Each band (a.k.a Landau level), contributes with Chern number +1

Hall conductivity in an insulator

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$

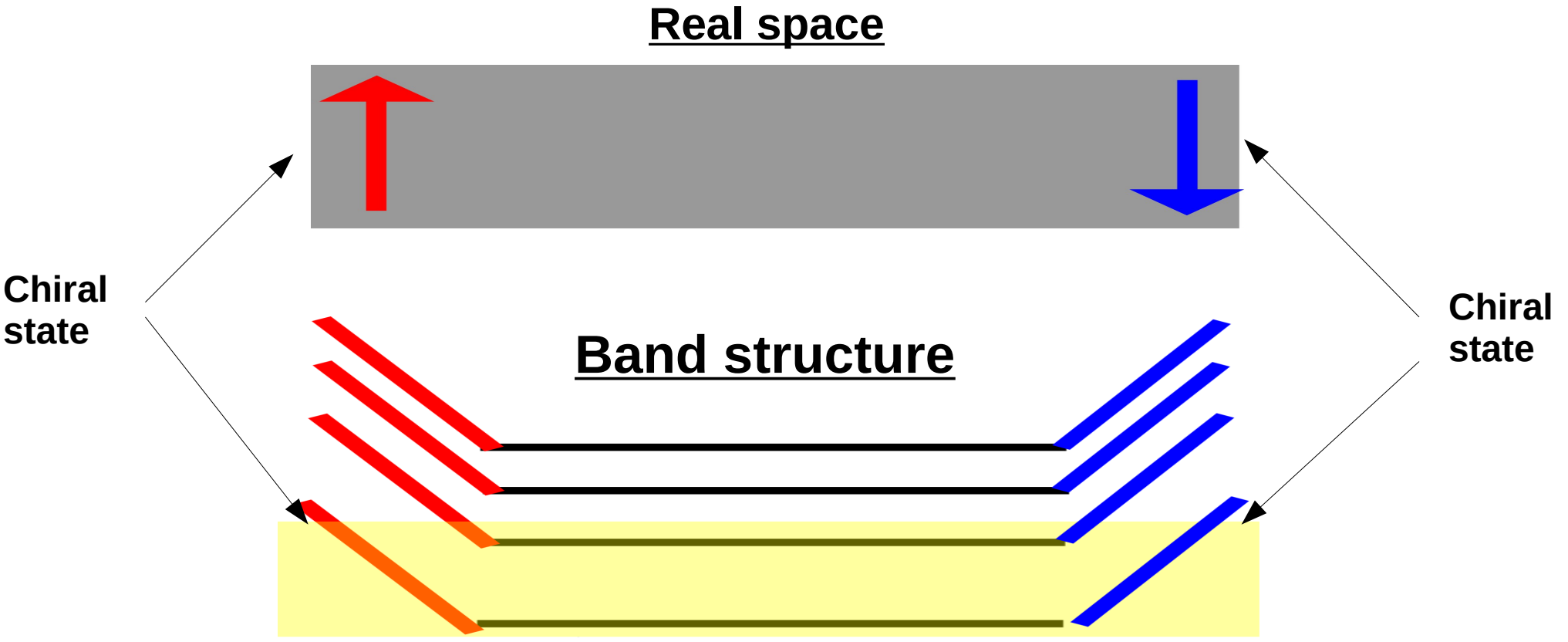
The Chern number for each band is quantized $C_\alpha = \int \Omega_\alpha(\mathbf{k}) d^2\mathbf{k} = 0, \pm 1, \pm 2, ...$



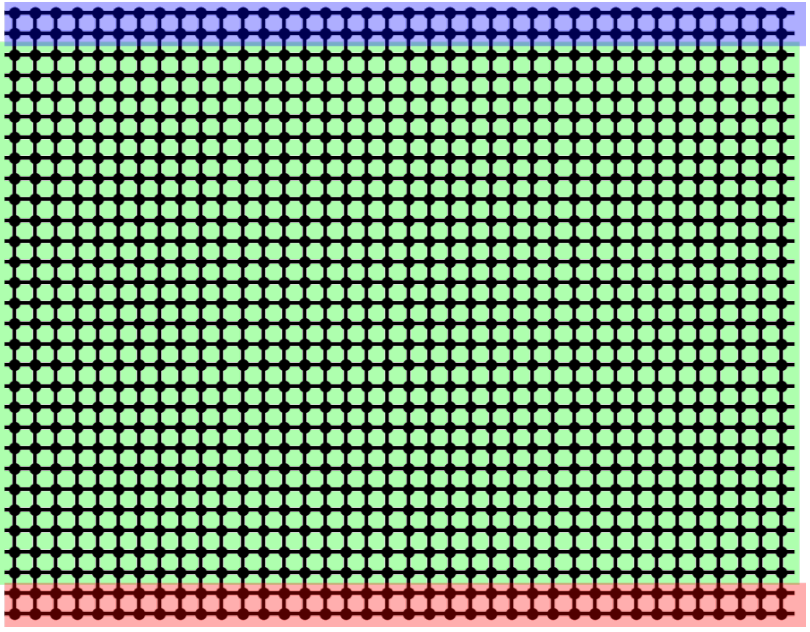
An insulator can have a finite (and quantized) Hall conductivity

This is a simple example of a topological state of matter

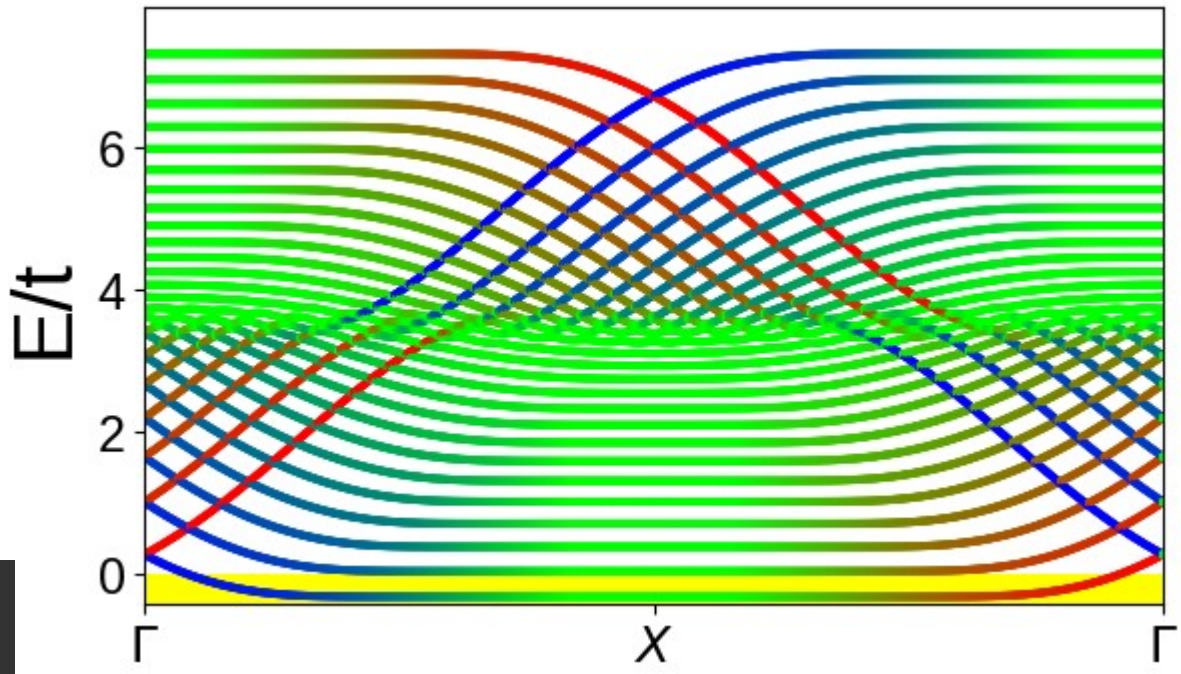
The quantum Hall effect



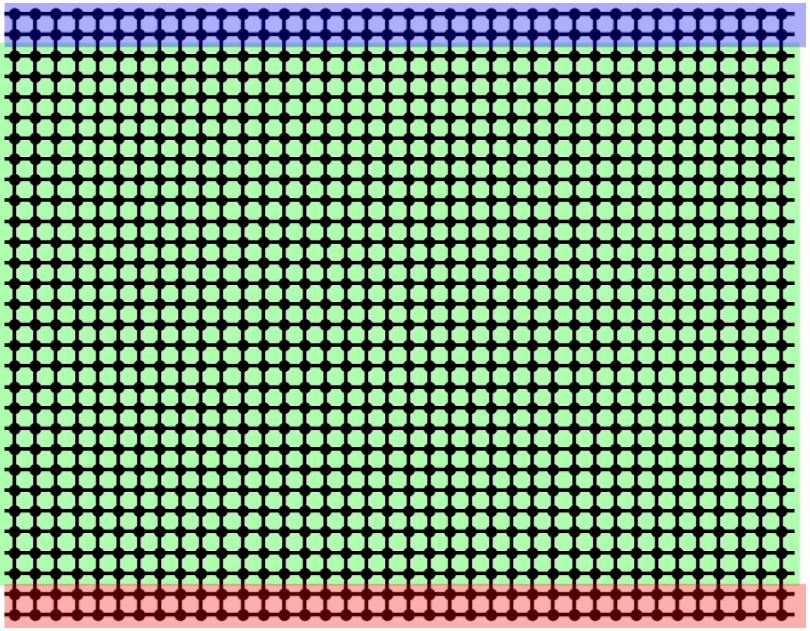
The quantum Hall effect in a square lattice



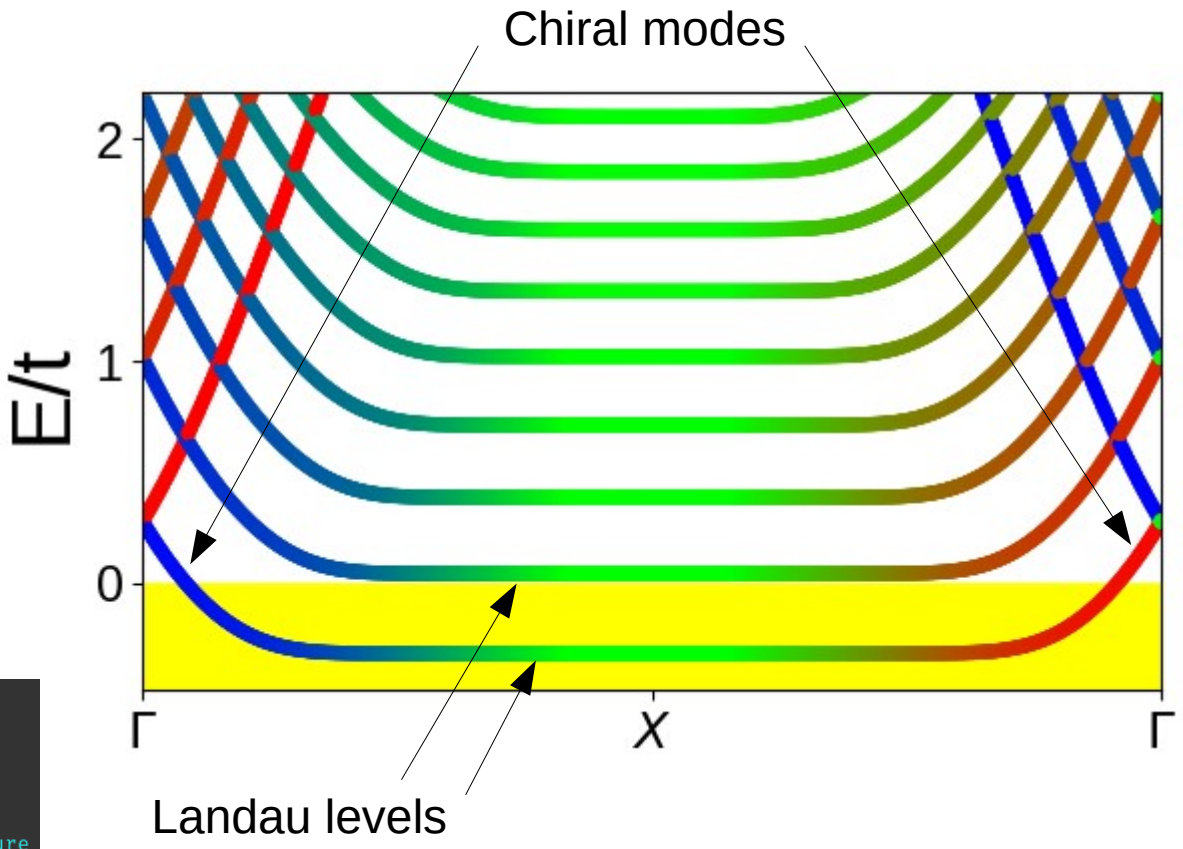
```
from pyqula import geometry,ribbon
g = geometry.square_lattice() # create the lattice
g = ribbon.bulk2ribbon(g,n=30) # create a ribbon
h = g.get_hamiltonian() # get the tight binding Hamiltonian
h.add_orbital_magnetic_field(0.03) # add magnetic field
h.add_onsite(3.5) # shift Fermi energy
(k,e,c) = h.get_bands(operator="yposition") # get the band structure
```



The quantum Hall effect in a square lattice



```
from pyqula import geometry,ribbon
g = geometry.square_lattice() # create the lattice
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```



Topology in electronic systems

The idea of topology

Holes in a 3d surface

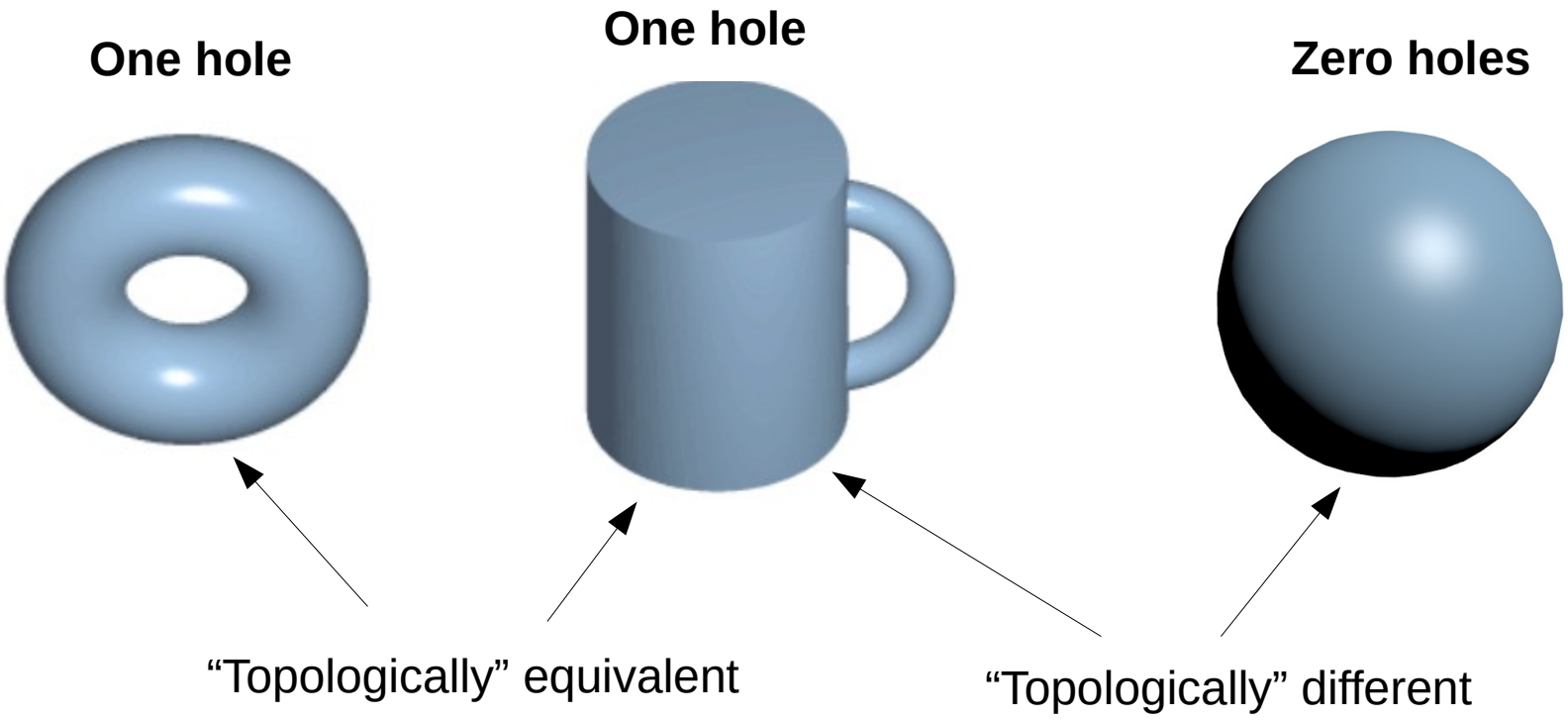


Knots in curves



Topology classifies object that cannot smoothly deformed into one another

Topology and holes



Topological invariant in a Hamiltonian

We can classify Hamiltonians according to topological invariants

Hamiltonian

$$H(\mathbf{k})$$

Wavefunction

$$|\Psi(\mathbf{k})\rangle$$

$$\Omega = \nabla \times \mathbf{A}|_z$$

Metric (Berry curvature)

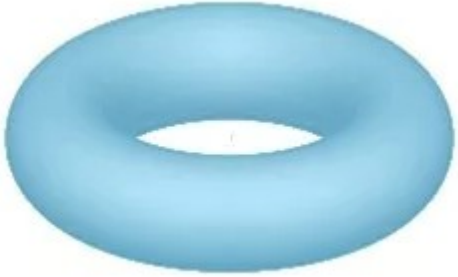
$$C = \int_{BZ} \Omega d^2\mathbf{k}$$

Topological invariant (Chern number)

The role of a topological invariant

Hamiltonians with different topological invariants can not be deformed one to another

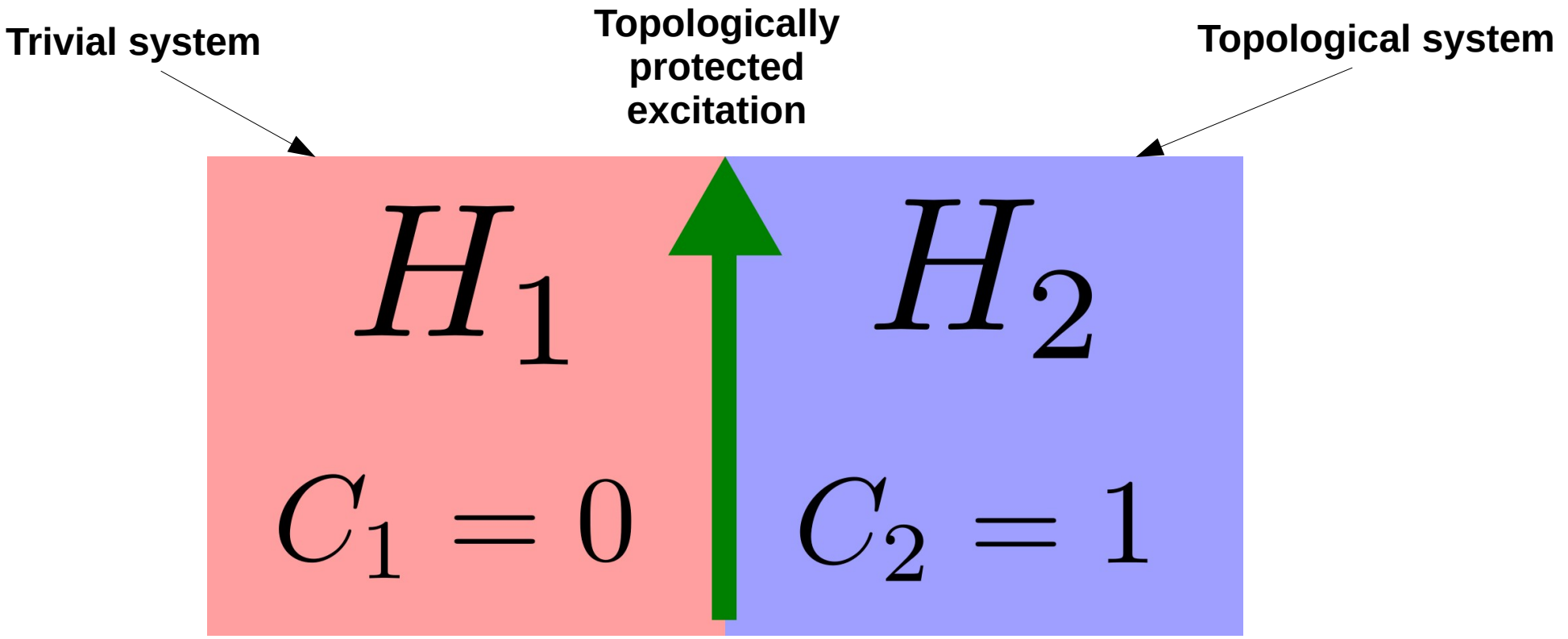
$C = 1$



$C = 2$

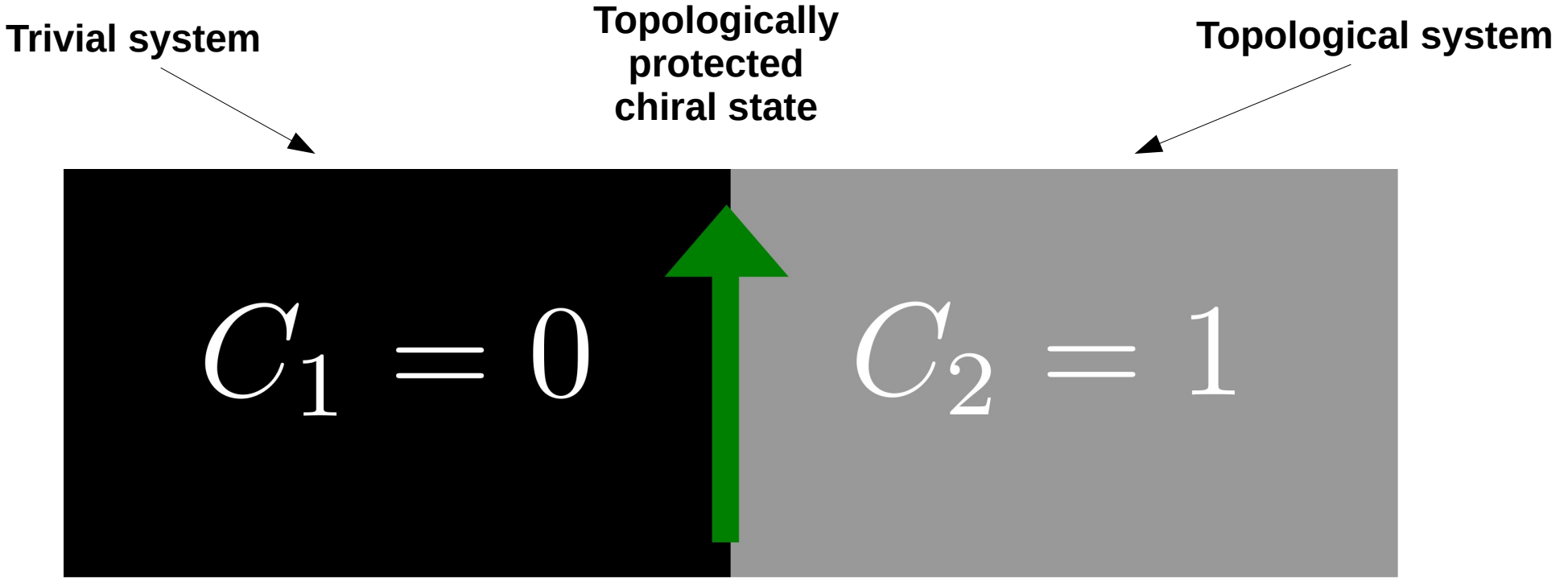


The consequence of different topological invariants



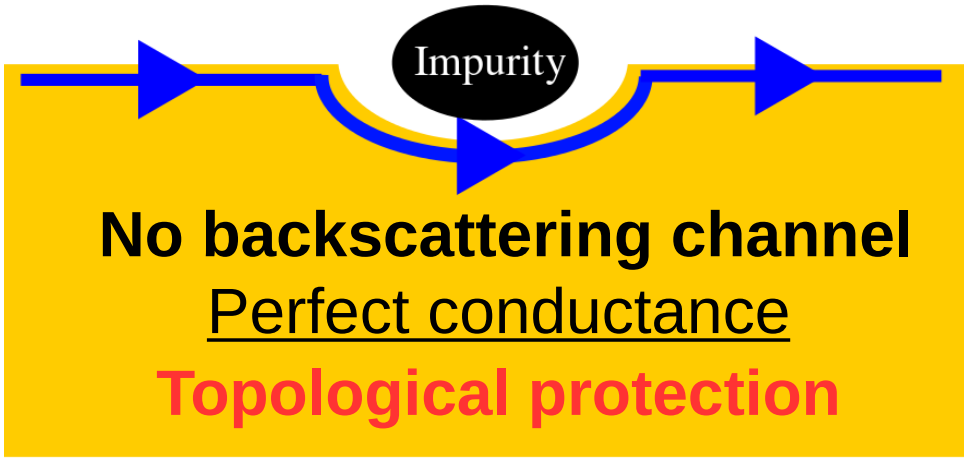
Topological excitations appear between topologically different systems

The edge states of the quantum Hall effect



The edge states of the quantum Hall effect are topological excitations

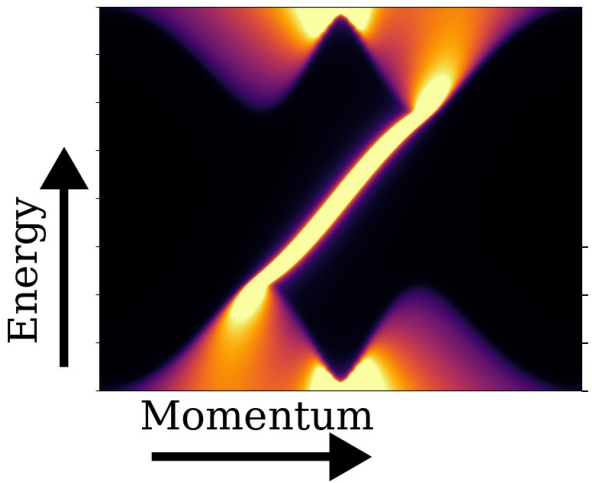
The edge states of the quantum Hall effect



The edge states of the quantum Hall effect are topologically protected

Three important topological materials

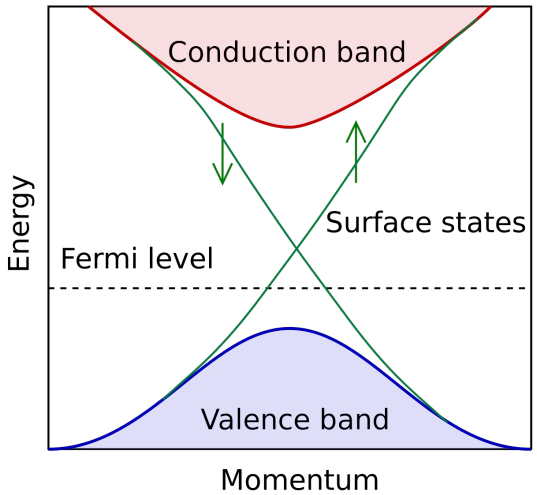
Chern insulators



Chiral states

Electronics

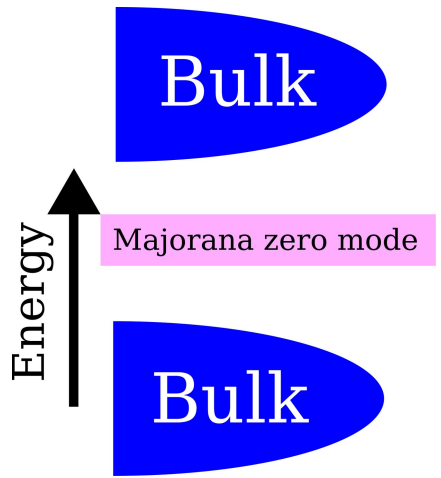
Quantum spin Hall insulators



Helical states

Spintronics

Topological superconductors

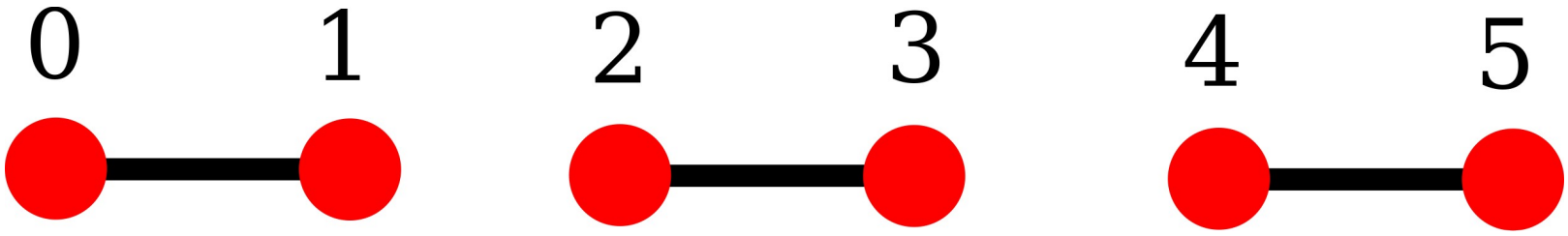


Majorana excitations

Topological quantum computing

A toy model for a topological insulator: the SSH model

The SSH model

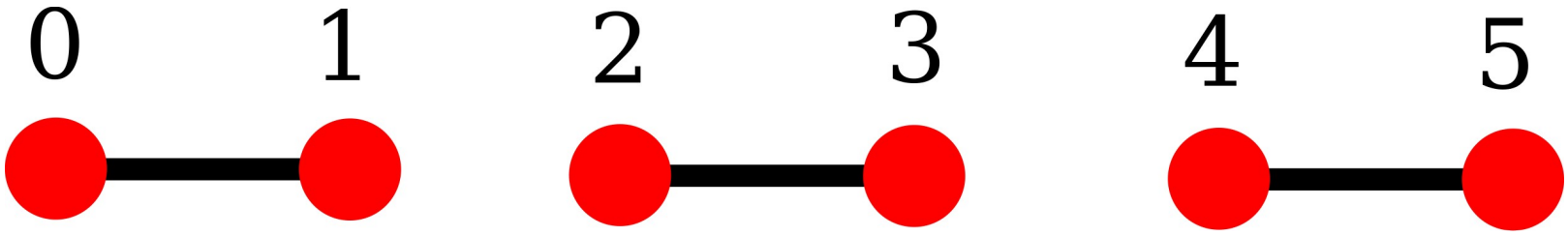


$$H = \sum_{n=0}^{4} [1 + (-1)^n]/2 c_n^\dagger c_{n+1} + h.c.$$

$$H = c_0^\dagger c_1 + c_2^\dagger c_3 + c_4^\dagger c_5 + h.c.$$

Let us consider a finite dimerized chain

The SSH model



$$H = \sum_{n=0}^{4} [1 + (-1)^n]/2 c_n^\dagger c_{n+1} + h.c.$$

What is the spectra of this Hamiltonian?

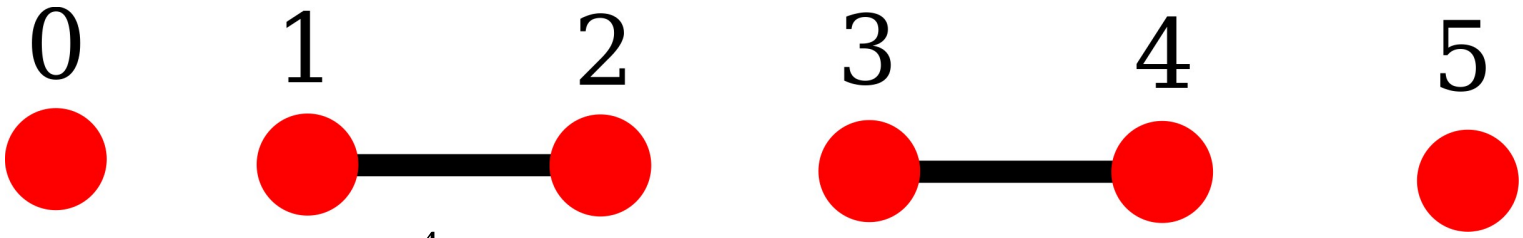
Option #1

$$\epsilon_n = \pm 1$$

Option #2

$$\epsilon_n = 0, \pm 1$$

The SSH model

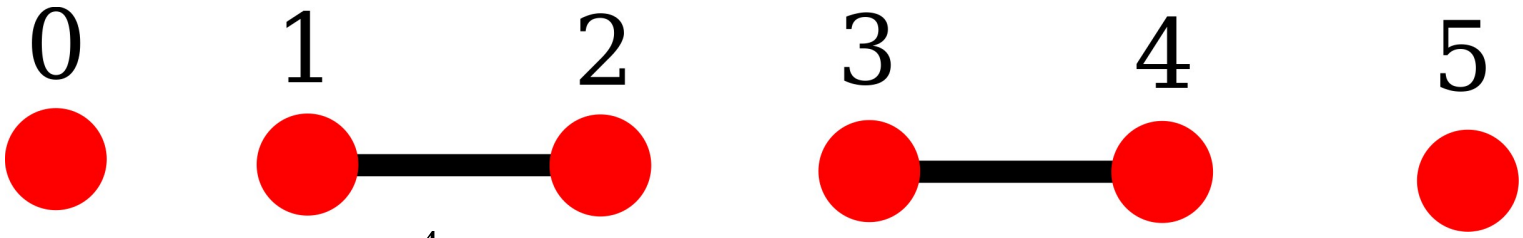


$$H = \sum_{n=0}^{4} [1 - (-1)^n]/2 c_n^\dagger c_{n+1} + h.c.$$

$$H = c_1^\dagger c_2 + c_3^\dagger c_4 + h.c.$$

Let us consider now the other dimerization (with dangling sites)

The SSH model



$$H = \sum_{n=0}^{4} [1 - (-1)^n]/2 c_n^\dagger c_{n+1} + h.c.$$

What is the spectra of this Hamiltonian?

Option #1

$$\epsilon_n = \pm 1$$

Option #2

$$\epsilon_n = 0, \pm 1$$

The two phases of the SSH model

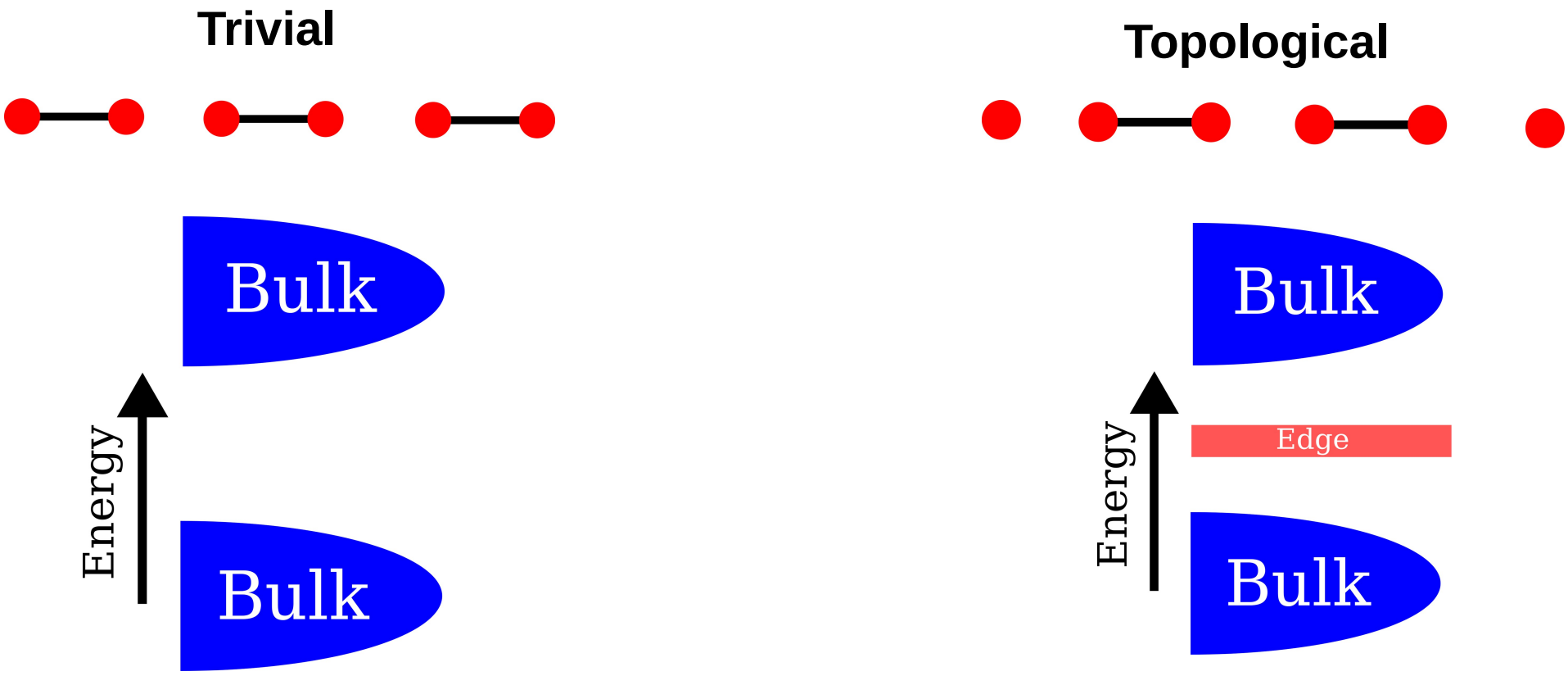
“Trivial” phase (gaped everywhere)



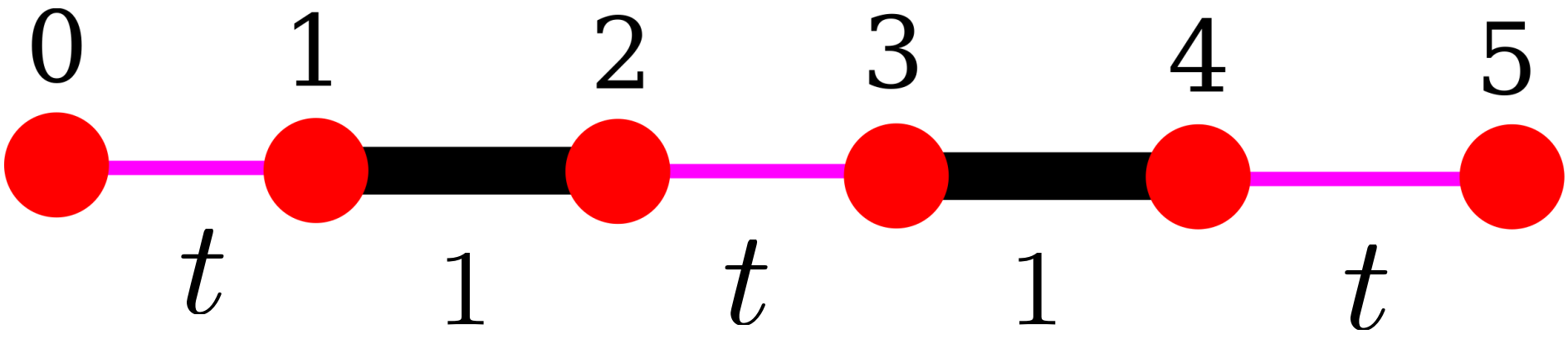
“Topological” phase (gapless zero modes)



The two phases of the SSH model



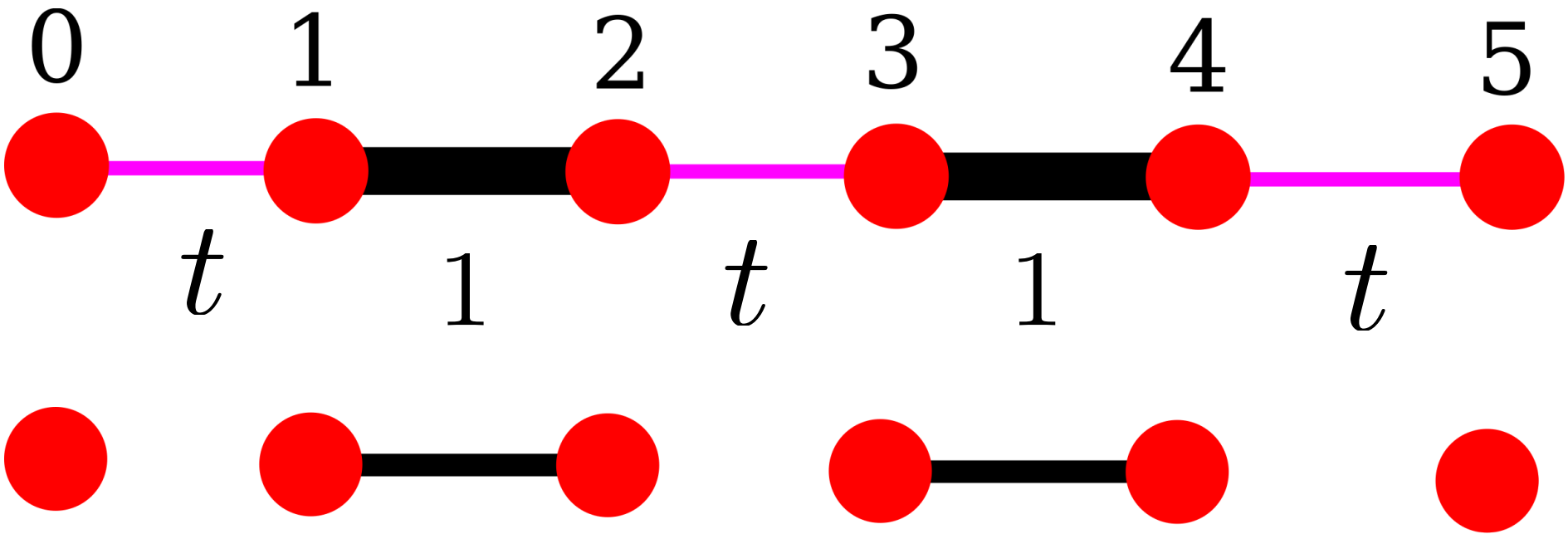
Coupling the dimers



$$H = tc_0^\dagger c_1 + c_1^\dagger c_2 + tc_2^\dagger c_3 + c_3^\dagger c_4 + h.c.$$

Does this Hamiltonian have a surface zero mode?

Coupling the dimers

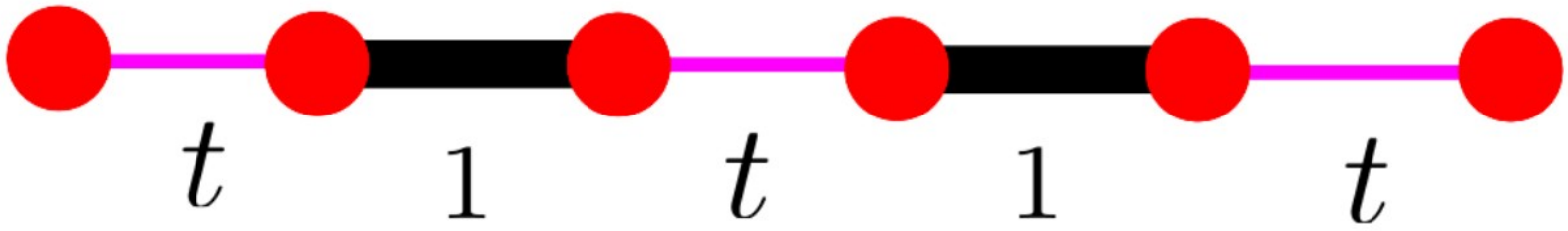


For t<1, both Hamiltonians are topologically equivalent

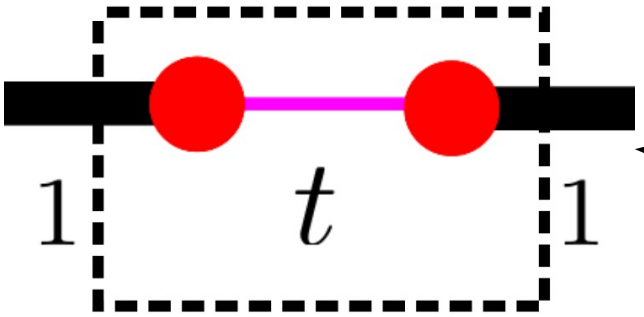
They can be deformed into one another without closing the bulk gap

The bulk Hamiltonian in the SSH model

For a finite system of this form



The unit cell is

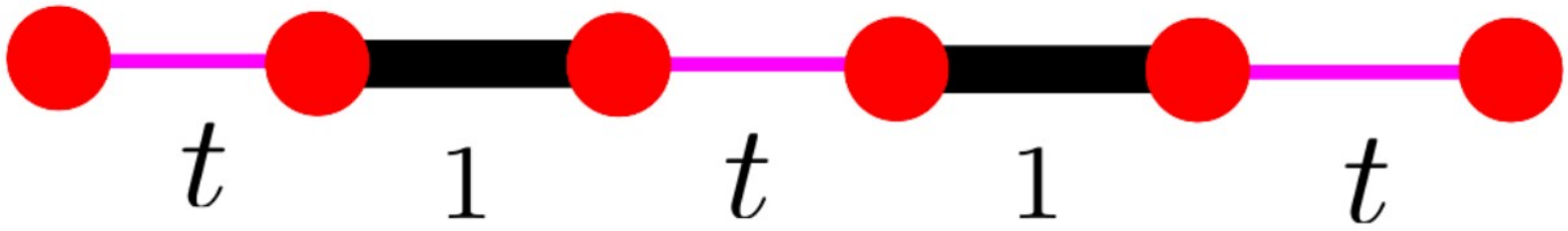


What is the Bloch Hamiltonian for this unit cell?

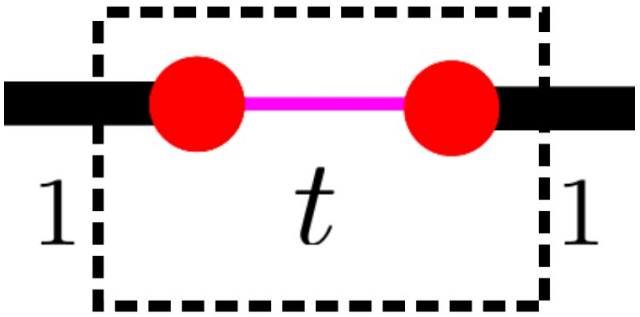
Hint: the Hamiltonian is a 2x2 matrix

The bulk Hamiltonian in the SSH model

For a finite system of this form



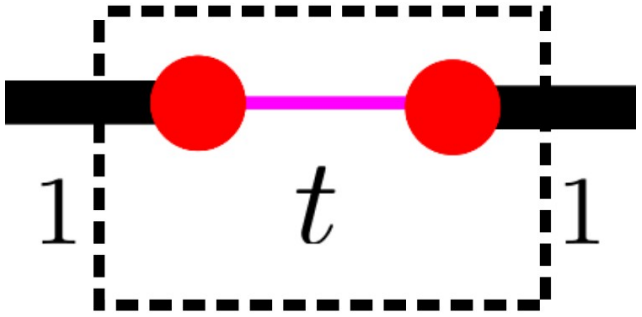
The unit cell is



The Hamiltonian is

$$H = \begin{pmatrix} 0 & t + e^{ik} \\ t + e^{-ik} & 0 \end{pmatrix}$$

The bulk invariant in the SSH model



The Hamiltonian is

$$H = \begin{pmatrix} 0 & t + e^{ik} \\ t + e^{-ik} & 0 \end{pmatrix}$$

$|\Psi(k)\rangle$ ← Lowest energy wavefunction

The topological invariant for this system is the Zak phase

$$\phi = \int_{BZ} A dk$$

$$A = i\langle \Psi(k) | \partial_k | \Psi(k) \rangle$$

Zak phase

The bulk invariant in the SSH model

Hamiltonian

$$H = \begin{pmatrix} 0 & t + e^{ik} \\ t + e^{-ik} & 0 \end{pmatrix}$$

Zak phase

$$\phi = \int_{BZ} A dk$$

$$A = i\langle \Psi(k) | \partial_k | \Psi(k) \rangle$$

Two different possible values for the Zak phase

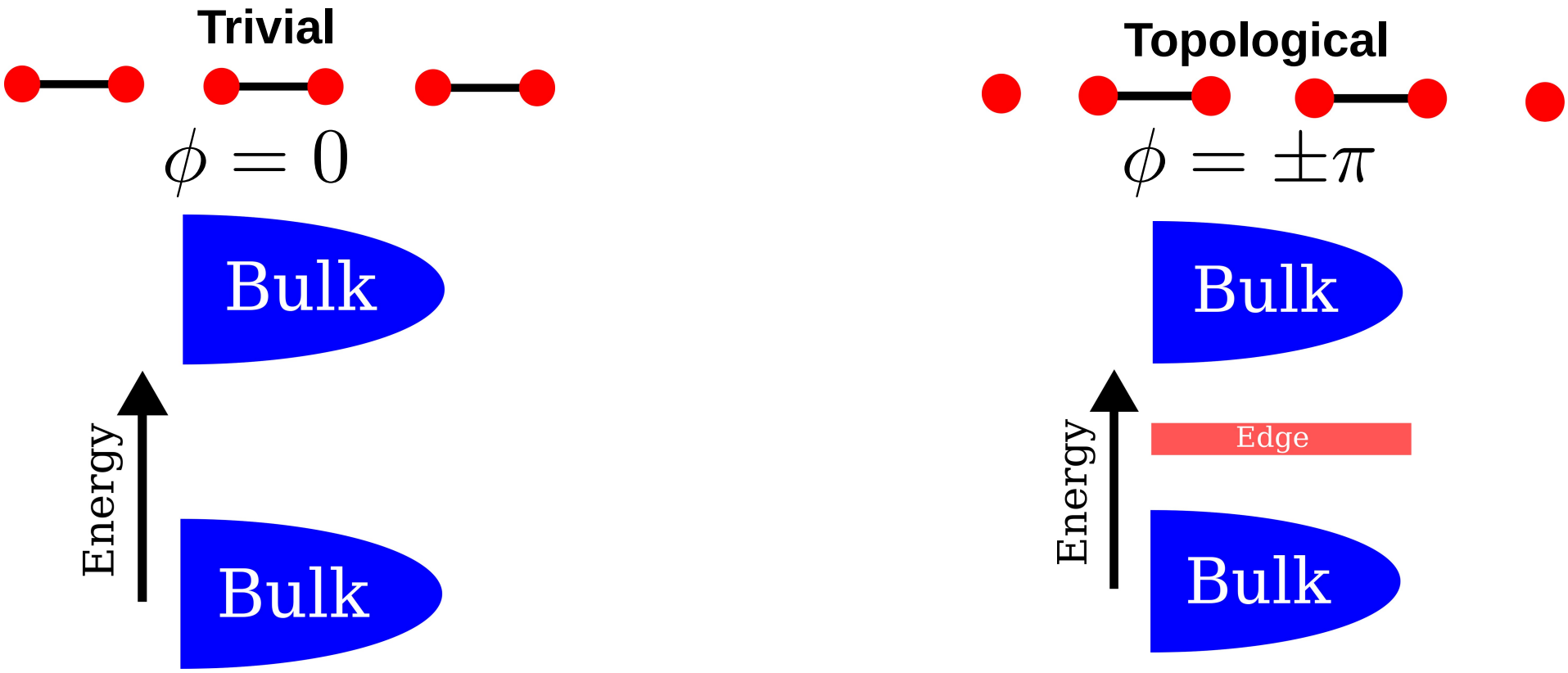
$$\phi = 0 \qquad t > 1$$

Trivial insulator

$$\phi = \pm\pi \qquad t < 1$$

Topological insulator

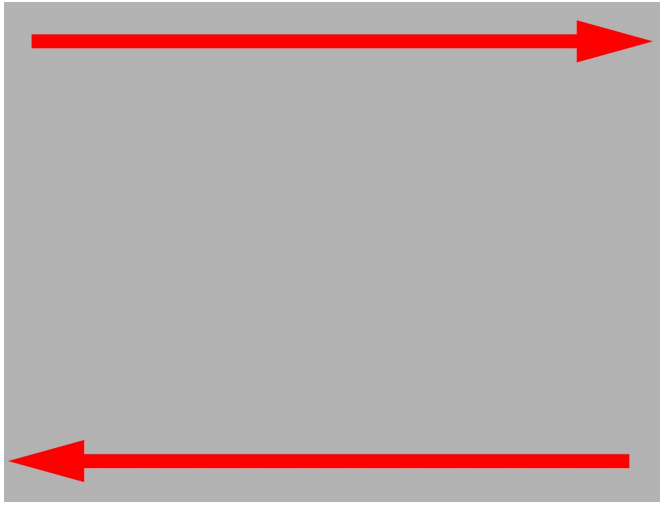
The bulk-boundary correspondence in the SSH model



Two-dimensional topological states

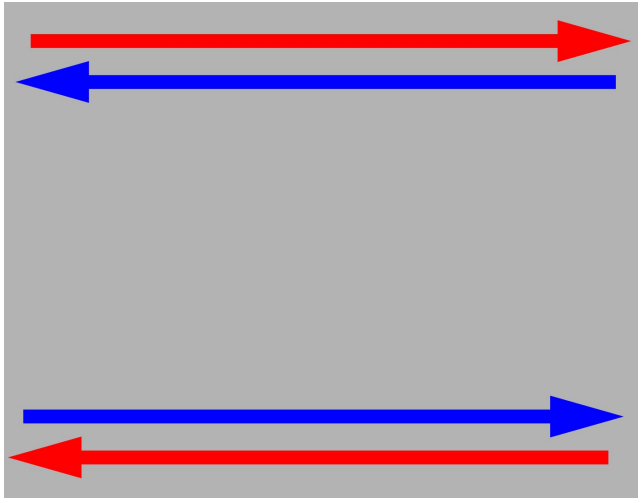
Two topological states in two dimensions

Chern insulators



Chiral modes
Break time-reversal symmetry

Quantum spin Hall insulators

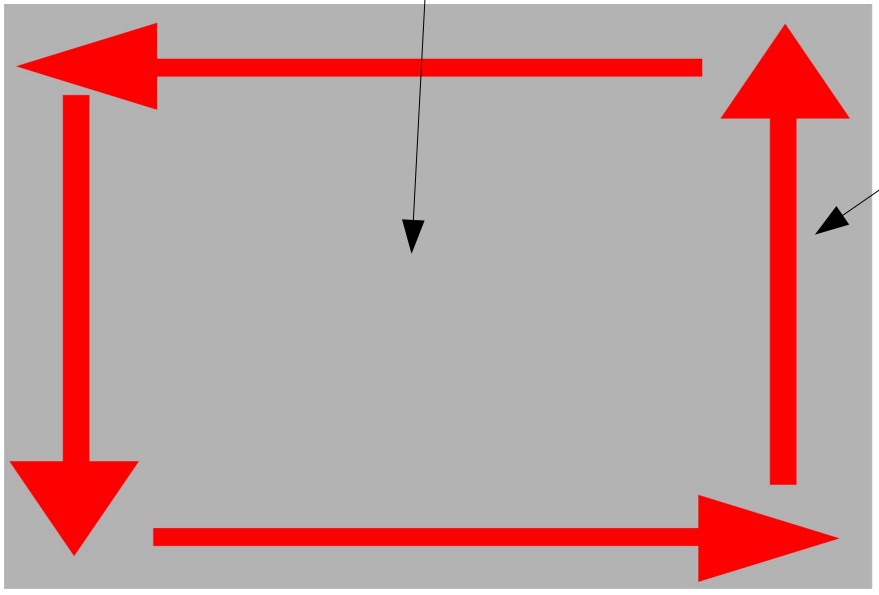


Helical modes
Do not break time-reversal symmetry

Chern insulators

Chern insulators

The bulk is insulating



The edge has chiral states (without an external magnetic field)

Hall conductivity (Chern number)

$$\sigma_{xy} = \sum_{\alpha \in occ} \int \Omega_\alpha d^2\mathbf{k} = \sum_\alpha C_\alpha = C$$

$$\Omega_\alpha = \partial_{k_x} A_y^\alpha - \partial_{k_y} A_x^\alpha$$

$$A_\mu^\alpha = i\langle \partial_{k_\mu} \Psi_\alpha | \Psi_\alpha \rangle$$

Building an artificial Chern insulator

How to design an electronic band structure with a finite Chern number?

Look for a system that has massive Dirac equations

$$H(p_x, p_y) = p_x\sigma_x + p_y\kappa\sigma_y + m\sigma_z = \begin{pmatrix} m & p_x + i\kappa p_y \\ p_x - i\kappa p_y & -m \end{pmatrix}$$

The finite mass gives rise to a Chern number $C = \frac{1}{2}\mathrm{sign}(m)\mathrm{sign}(\kappa)$

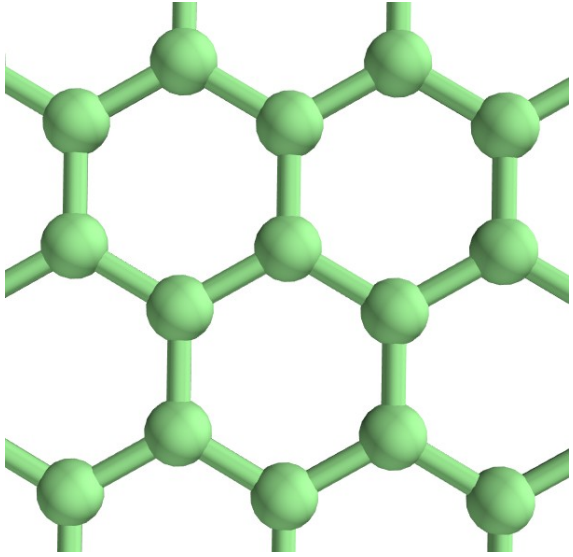
$$C = \int \Omega d^2\mathbf{k} \qquad \Omega = \partial_{k_x} A_y - \partial_{k_y} A_x \qquad A_\mu = i\langle \partial_{k_\mu} \Psi | \Psi \rangle$$

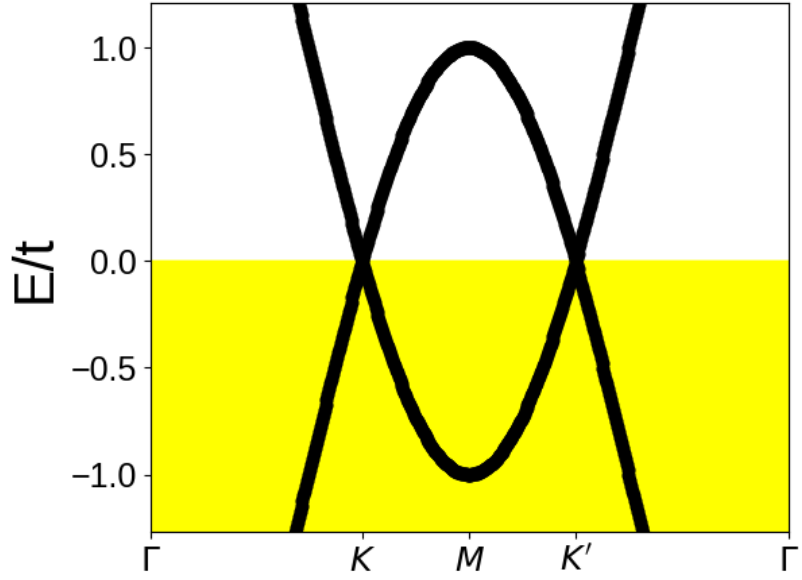
Which lattice model shows Dirac equations?

Building an artificial Chern insulator

How to design an electronic band structure with a finite Chern number?

Open up a gap in the Dirac points by breaking time-reversal symmetry



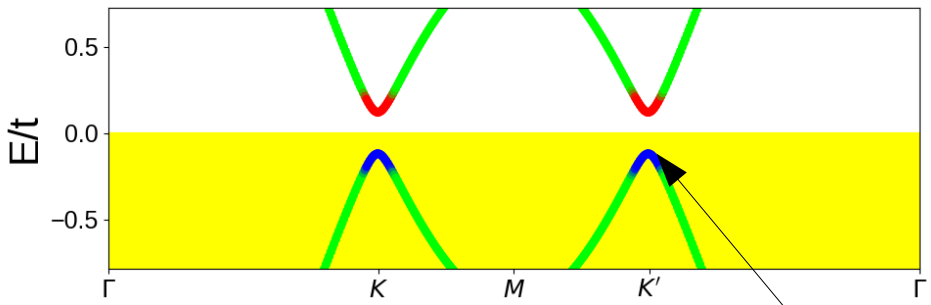


Building an artificial Chern insulator

How to design an electronic band structure with a finite Chern number?

Open up a gap in the Dirac points by breaking time-reversal symmetry

With internal orbital currents

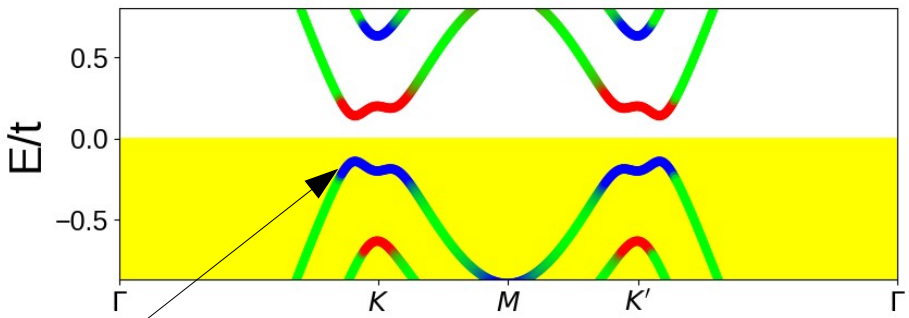


Haldane model

Realized in twisted graphene bilayers

Berry curvature

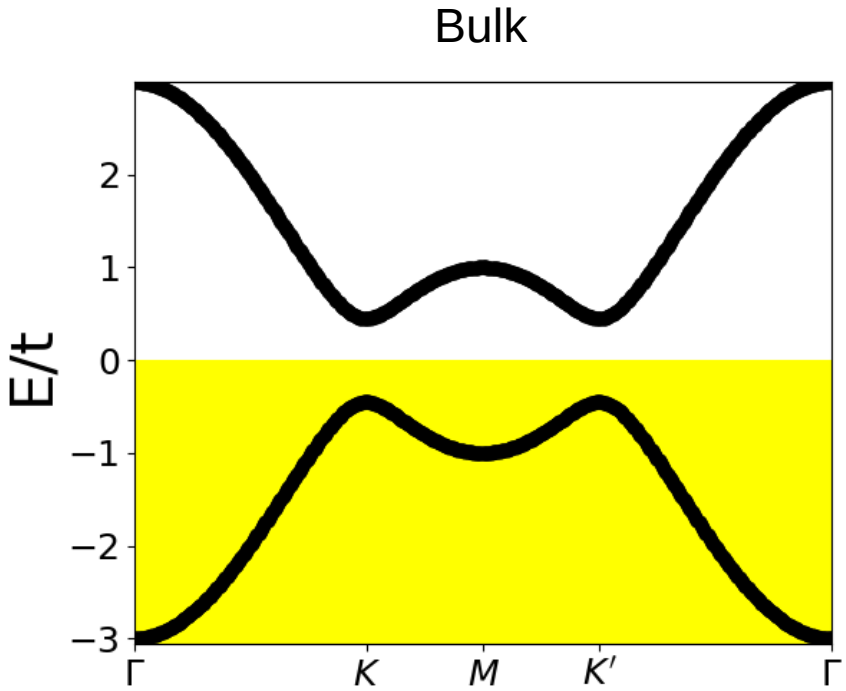
With magnetism and spin-orbit coupling

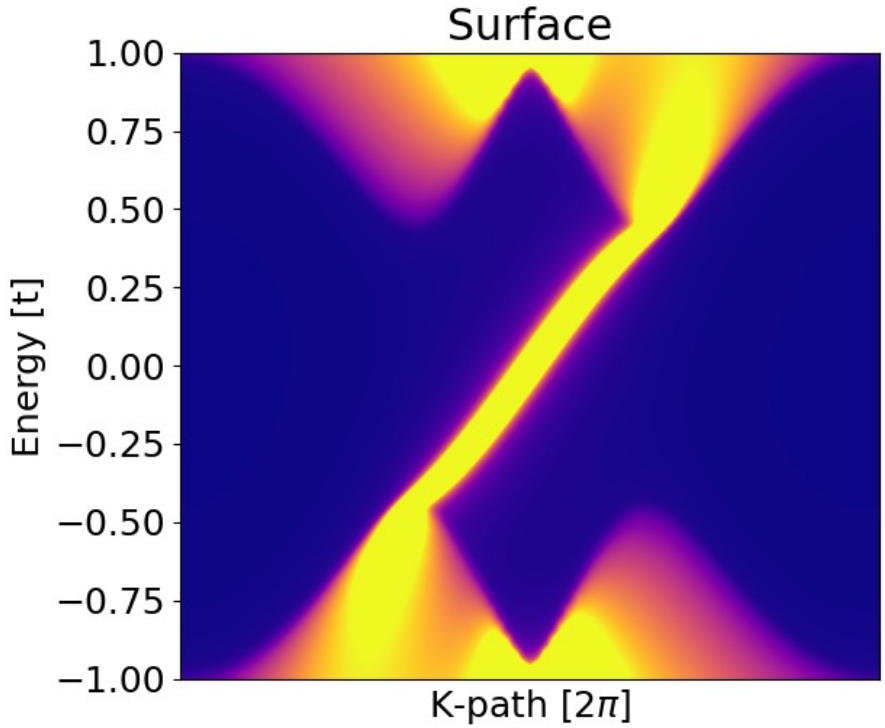


Magnetic helical gas

Realized in magnetically doped Bi_2Se_3

Bulk boundary correspondence in Chern insulators



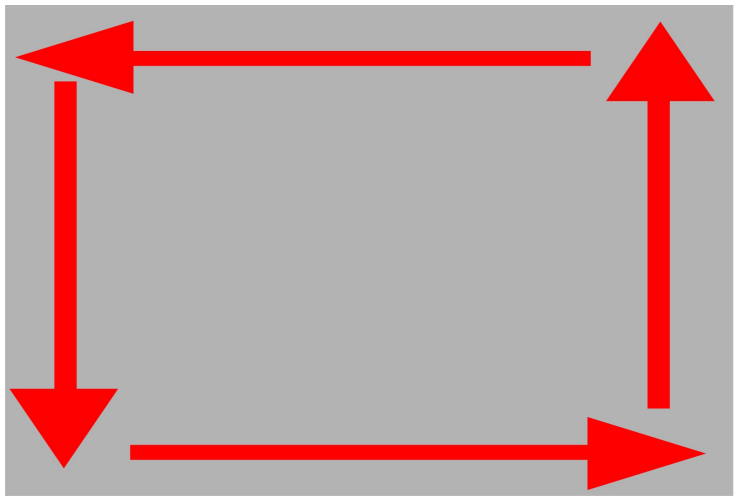


Quantum spin Hall insulators

Quantum spin Hall insulators

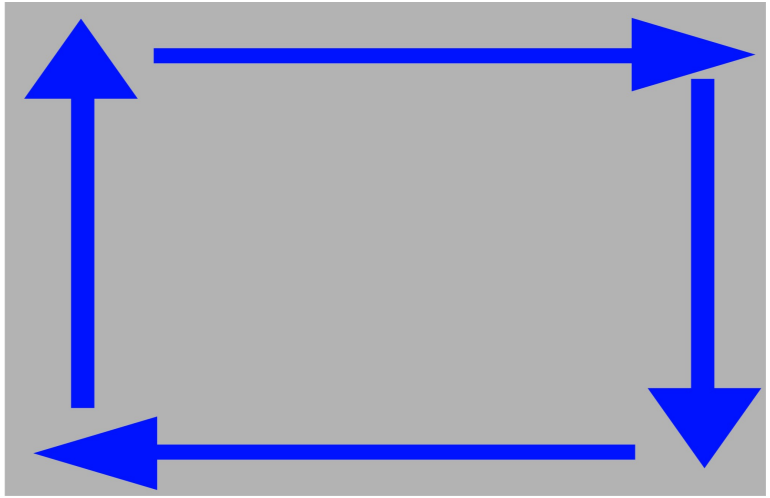
Chern insulator for spin up

$$C^{\uparrow} = 1$$



Chern insulator for spin down

$$C^{\downarrow} = -1$$

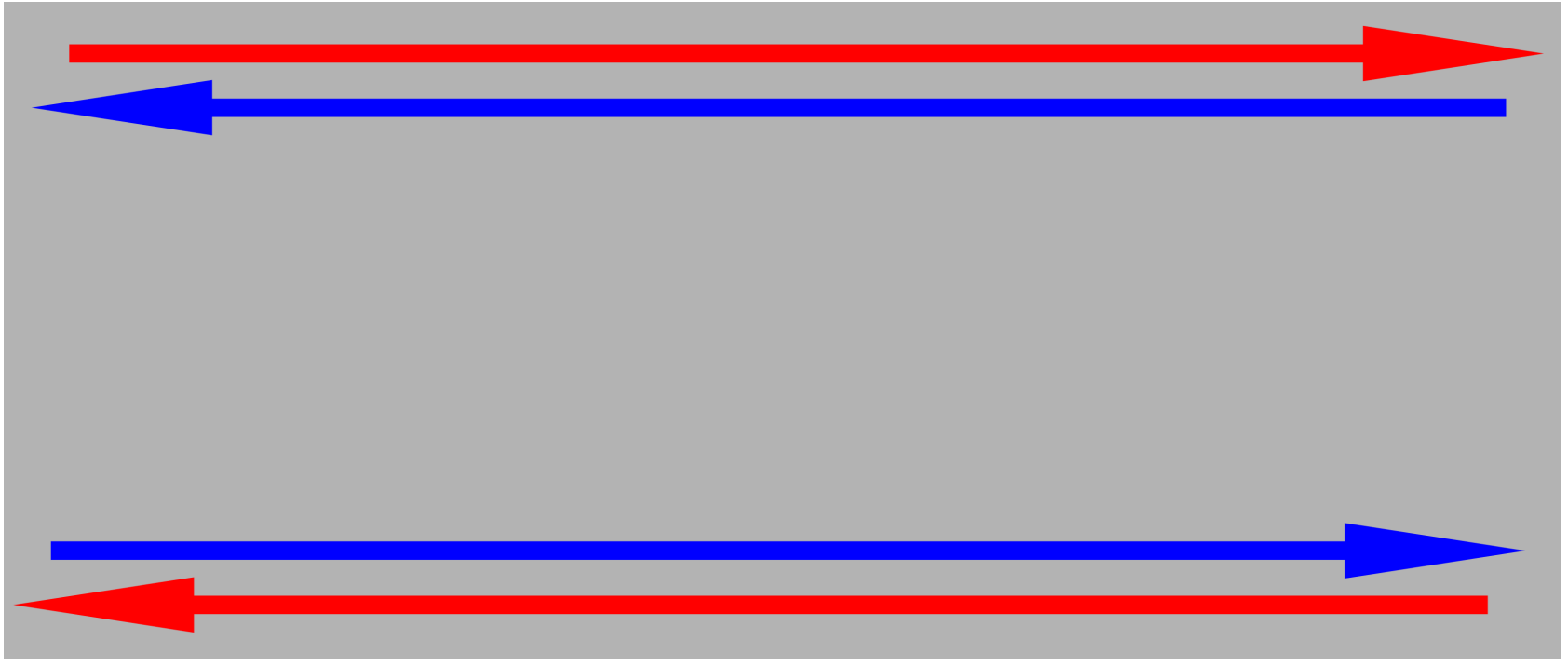


Spin-orbit coupling (SOC) can drive a quantum spin Hall state

$\vec{L} \cdot \vec{S} \sim L_z S_z$ *SOC acts as a magnetic field with opposite signs for opposite spins*

Quantum spin Hall insulators

Opposite spins propagate in opposite directions (helical gas)



The Hamiltonian of a topological insulator

Look for a system that has massive Dirac equations

$$H(p_x, p_y) = p_x\sigma_x + p_y\kappa\sigma_y + m\sigma_z = \begin{pmatrix} m & p_x + i\kappa p_y \\ p_x - i\kappa p_y & -m \end{pmatrix}$$

The finite mass gives rise to a local Chern number $C_{s,\alpha} = \frac{1}{2}\mathrm{sign}(m)\mathrm{sign}(\kappa)$

If the mass for spin up is opposite than for spin down, then

Chern number

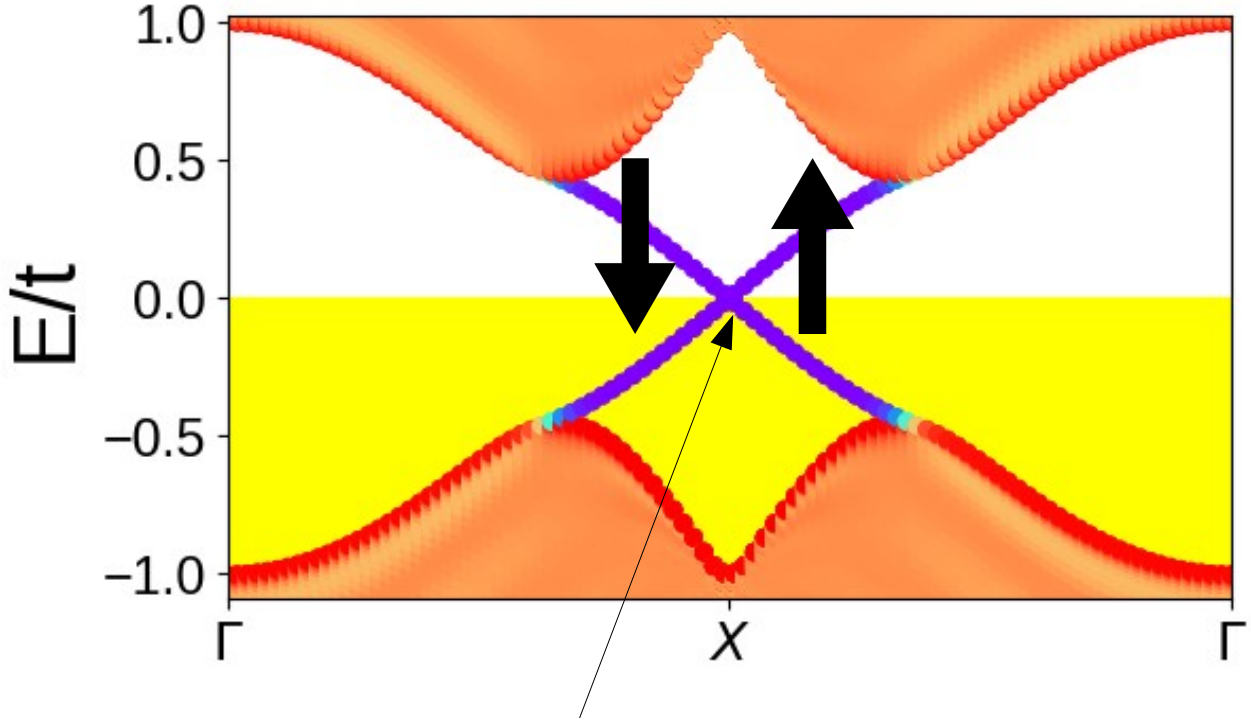
$$C = C_\uparrow + C_\downarrow = 0$$

Spin Chern number

$$C_S = C_\uparrow - C_\downarrow = \pm 2$$

In a system with spin-orbit coupling, spin dependent masses can be generated

Quantum spin Hall insulators



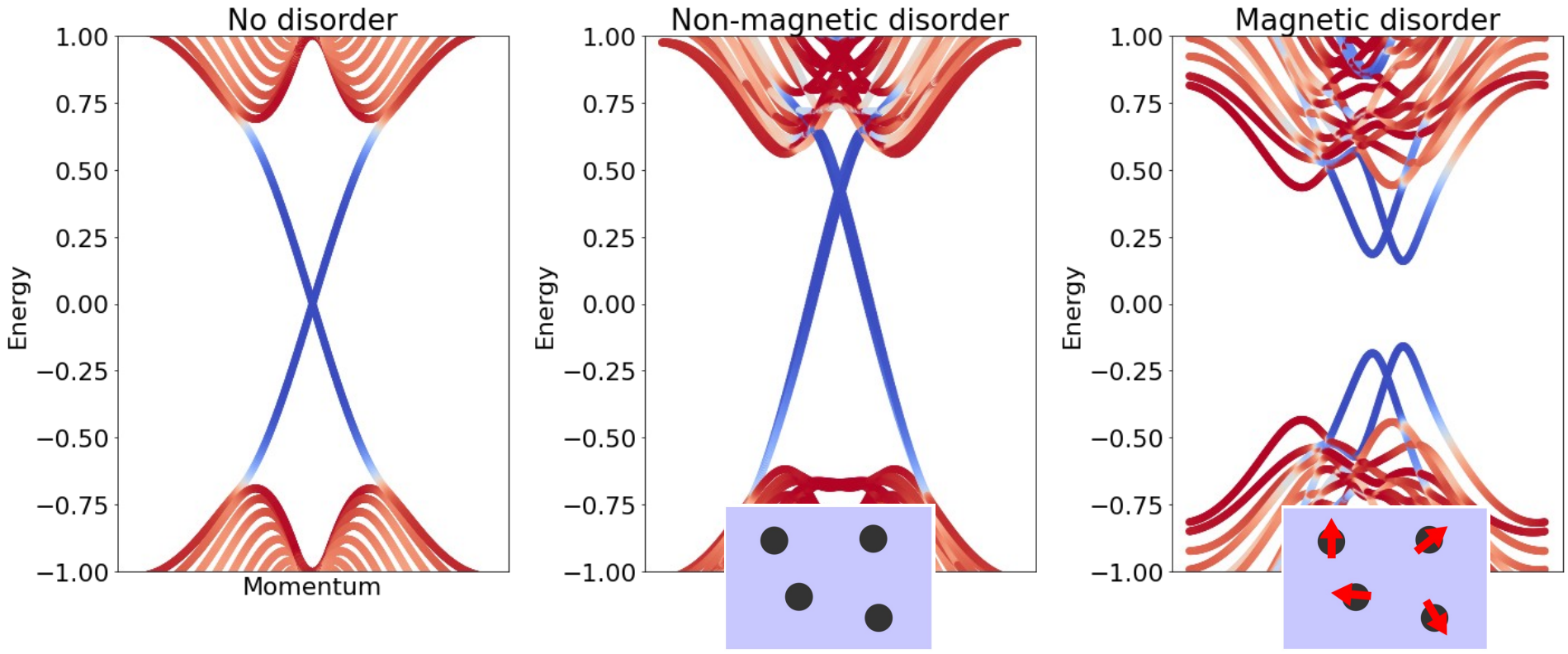
Time-reversal symmetry

$$\Theta^2 = -1$$

The gapless nature is protected by Kramer's theorem

Realized in 1T-WTe_2

Disorder in quantum spin Hall insulators



Disorder that breaks time reversal symmetry opens a gap in the topological states

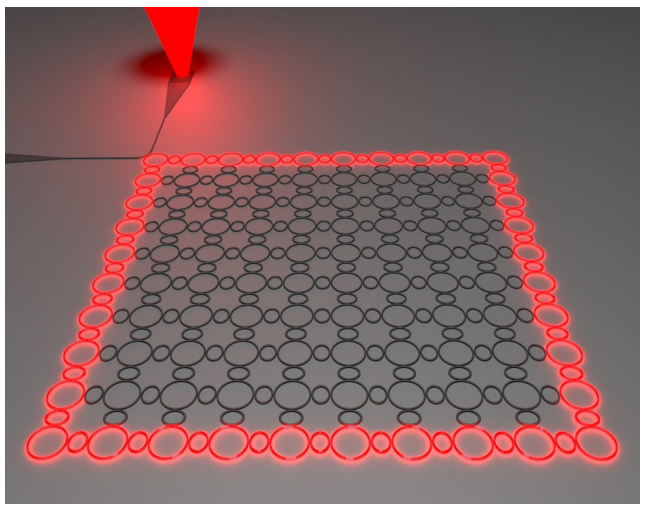
Topology beyond electrons

Topological modes beyond electrons

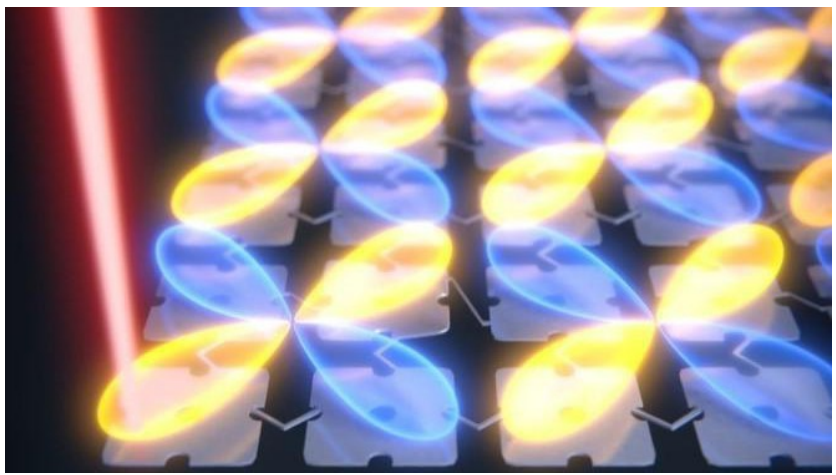
Topological edge modes are a signature of systems that can be described with matrices

Other systems can be described with mathematically analogous tools

Photonic systems



Classical (mechanical) systems



Take home

- Topological systems have protected edge modes
- The existence of edge modes is associated with a non-trivial topological invariant